

3rd CII National Kaizen Circle Compétition 2020

A Kaizen Presentation By

(Everest Industries Ltd)

Podanur Works - Coimbatore

24th ~ 26th June 2020

COMPANY OVERVIEW

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

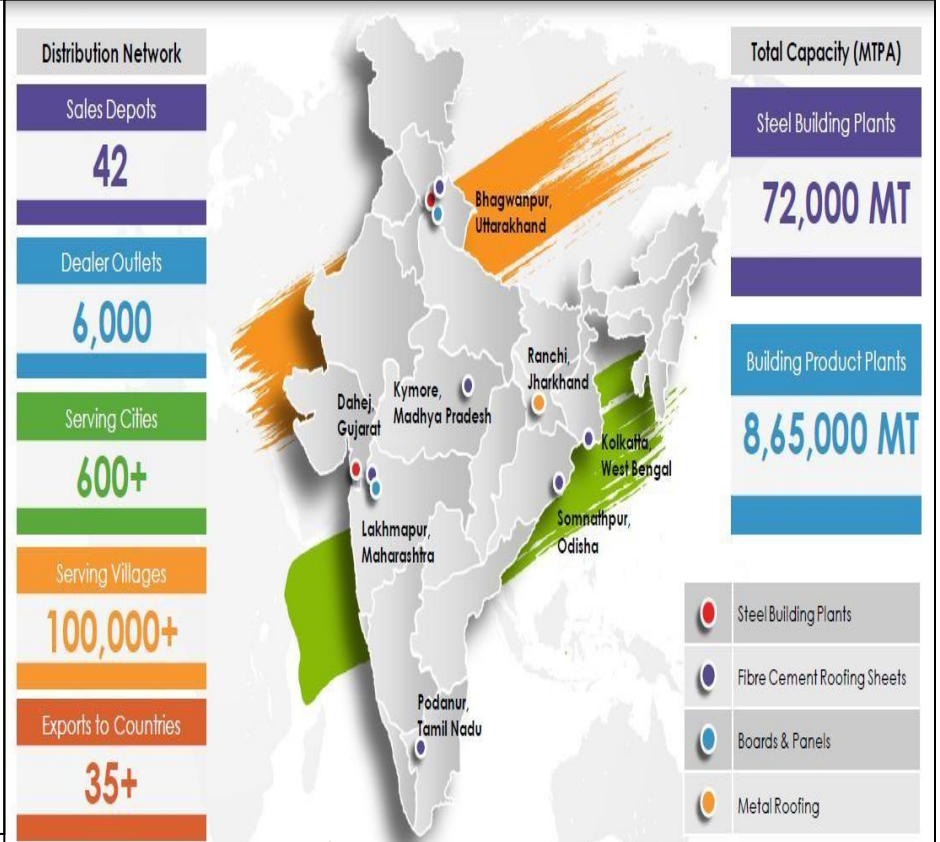
OUR BUSINESS

The company businesses are divided into two distinct business segments with capacity –

1. Buildings Products – 8,65,000 MT / Annum
2. Steel Buildings – 72,000 MT / Annum

Buildings Products segment contributed 63% of the company's revenue and includes Fiber cement roofing sheets, Fiber cement boards, Rapicon wall & their accessories.

Steel Building segment accounted for 37% of annual revenue. In this segment Company offers customized building solutions in steel such as Pre-Engineered steel Buildings, Smart steel buildings, Metal roofing and Cladding.



Products:



Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings

Customers:



Kaizen Theme: To reduce Green sheets loss and recycling by reducing Piling Bogie changeover time.

Implemented Area: Line No -1, Podanur Works, Coimbatore

Implementation Date: 12th October 2019

Cross Functional Team:



Mr. Y. Karthikeyan
(Sr.Engineer - Maintenance)



Team Leader
Mr. C. Murugesan
(Maintenance Manager)



Mr. M. Vijay Kumar
(Sr.Engineer - Electrical)



Mr. Y. Kesavaraj
(Sr.Engineer - Line In-charge)



Mr. M. Madhan Raj
(Prod. Shift Incharge)

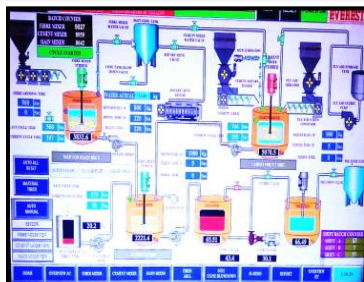


Mr. N. Anil Kumar Reddy
(Engineer - QS & BE)

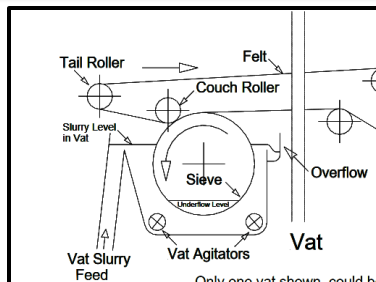


Mr. Devaraj
(Fitter)

Process Flow:



1) RAW MATERIAL PREPARATION



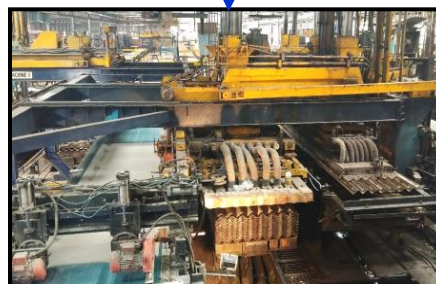
2) VATS



3) SHEET FORMING



7) EMPTY BOGIE IN PILING M/C



4) SHEET HANDLING / PILING MACHINE



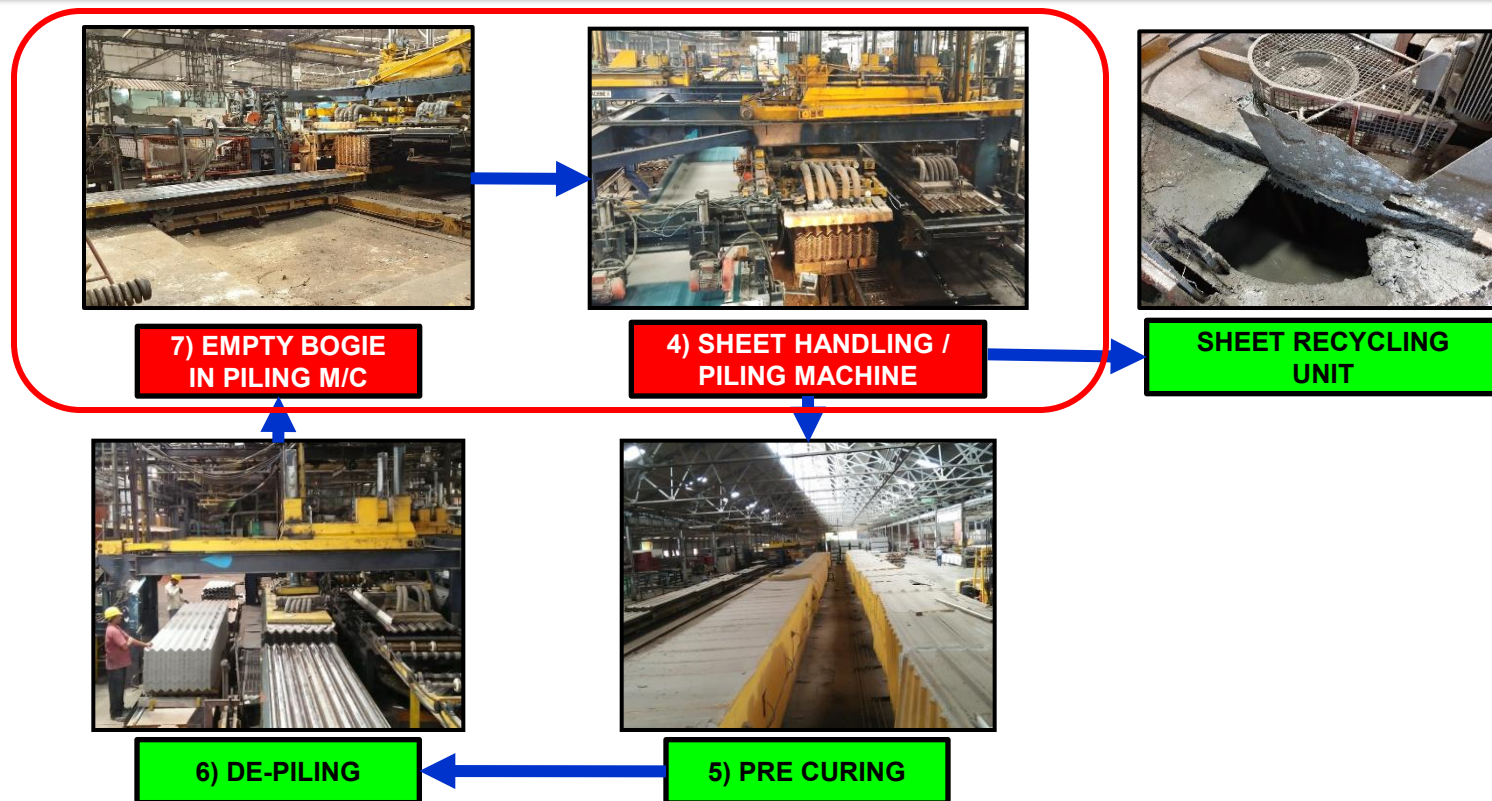
SHEET RECYCLING UNIT



6) DE-PILING



5) PRE CURING

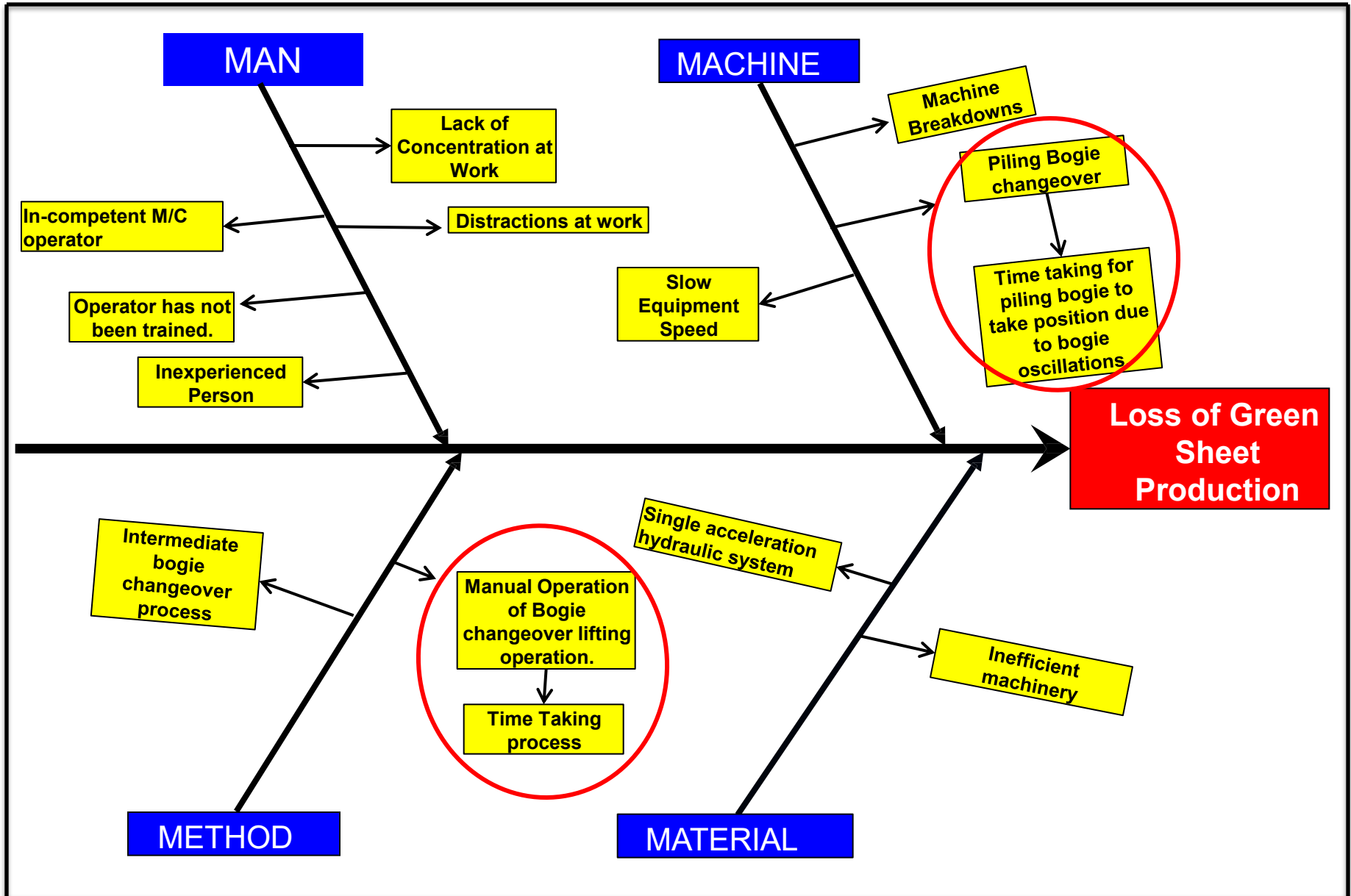


Problem/ Present Status :

Green Sheets loss during piling bogie changeover. As there is time delay of empty bogie to take it's position in Piling machine.

- ❖ For Piling Bogie changeover, it takes 45 sec to complete the changeover. And Sheet formation time is 16 sec. By this we are losing 3 Green sheets to recycling per piling bogie changeover.

Root Cause Analysis



Problem Statement	Loss of Green sheets production.
Why 1	Due to piling bogie changeover.
Why 2	More time taking i.e. 45 sec for piling bogie changeover.
Why 3	When new piling bogie reached it's position, it oscillates back & front. Due to this there is delay in lifting the bogie.
Why 4	There is no mechanical stopper to stop the bogie oscillations immediately after bogie reaches it's position.

Root Cause:

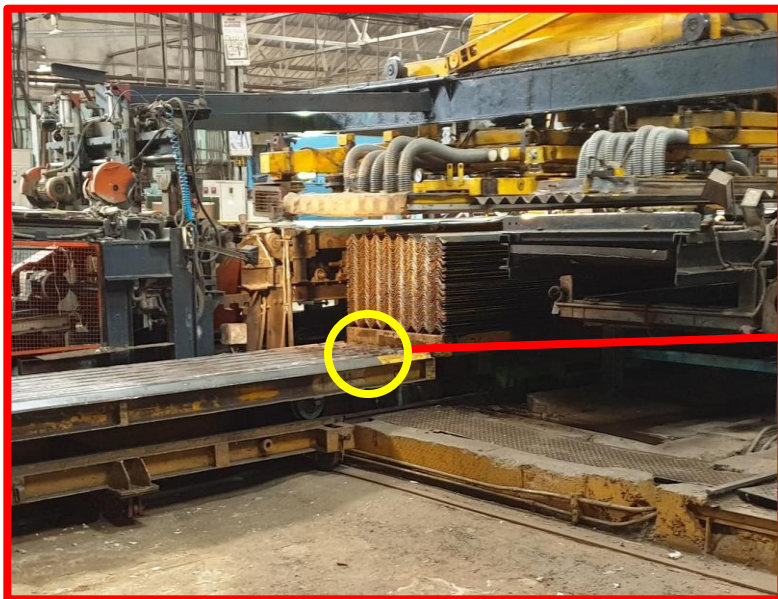
No Mechanical stopper arrangement to restrict the bogie oscillations immediately after the bogie reaches it's position.

Idea Generation: To provide Locking arrangement for restricting the bogie oscillations immediately.

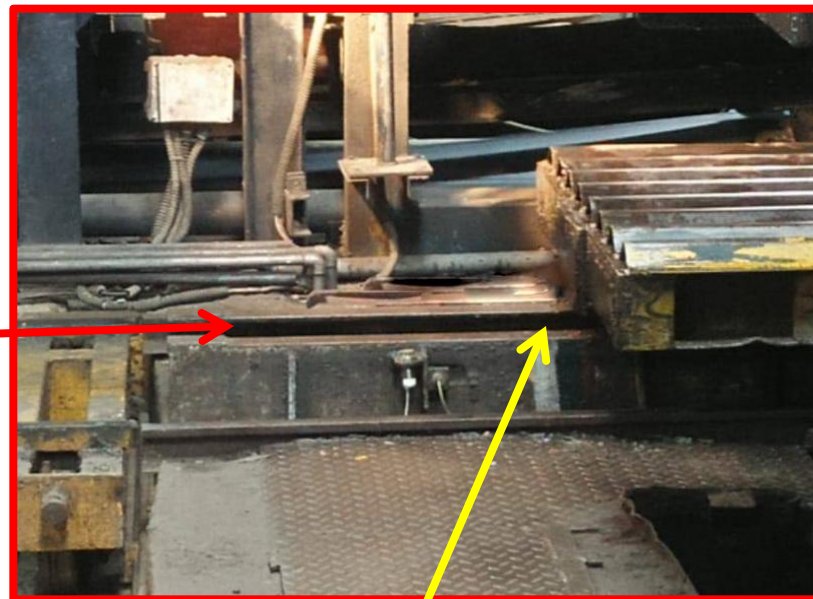
Action:

To provide Mechanical lock for restricting the bogie oscillations immediately when new piling bogie takes it's position in piling machine. And to provide PLC automation for lifting bogie.

S.No	Counter Measure Activities	Start Date	Target Date	Responsibility	Status
01	To study design & process of piling machine.	05/10/2019	08/10/2019	Production & Mechanical Department	Completed
02	To provide self actuation mechanical lock.	09/10/2020	11/10/2020	Mechanical Department	Completed
03	To provide automation in PLC for lifting bogie.	11/10/2020	12/10/2020	Electrical Department	Completed

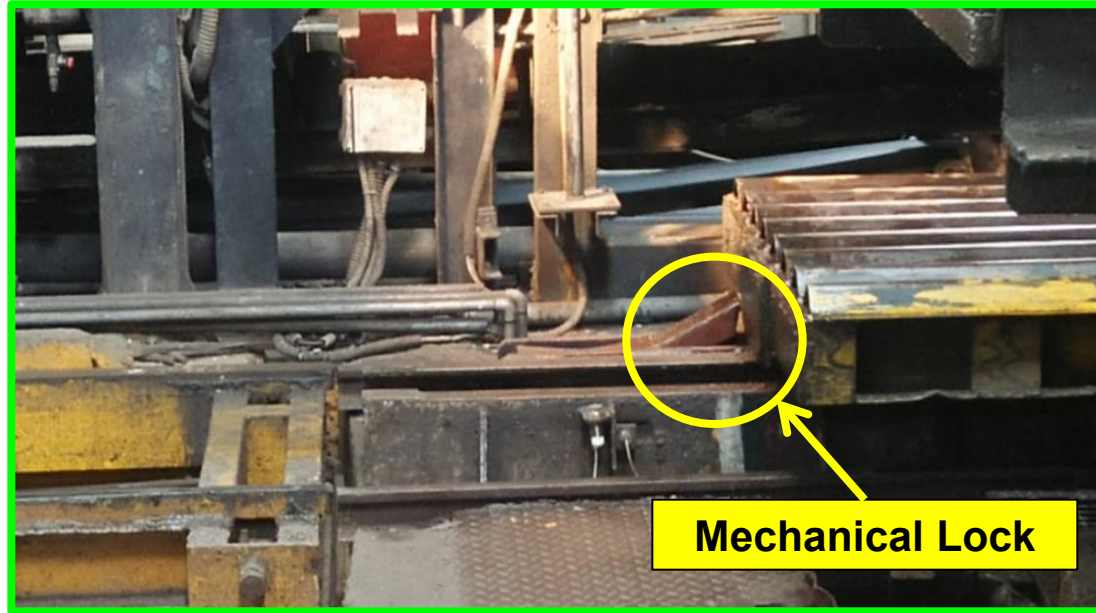


Empty bogie & filled bogie in piling M/C



No Mechanical Lock for restricting bogie oscillations.

- ❖ For Piling Bogie changeover, it takes 45 sec to complete the changeover. And Sheet formation time is 16 sec. By this we are losing 3 Green sheets to recycling per piling bogie changeover.

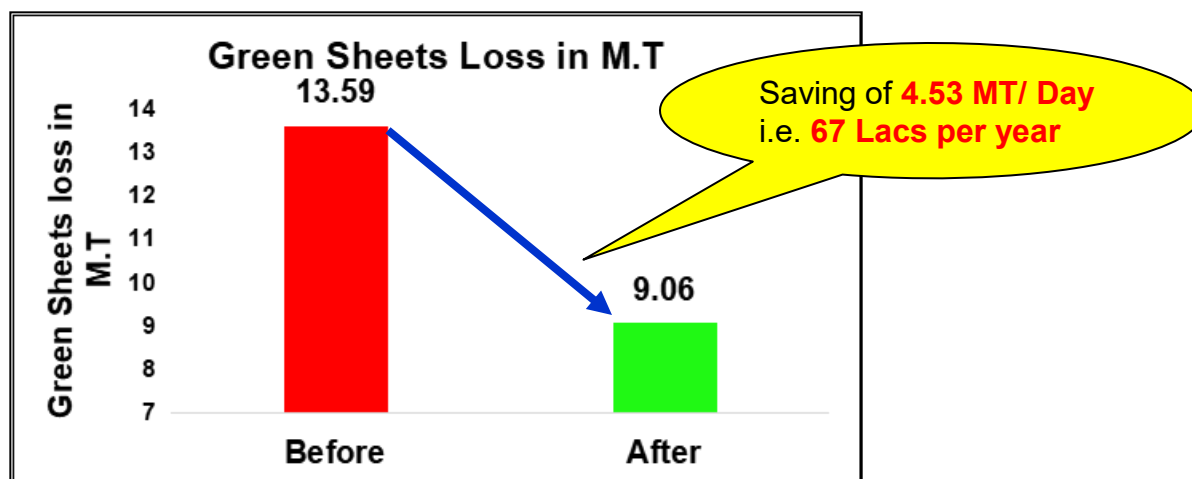


- ❖ Mechanical lock provided for restricting the bogie oscillations. And bogie changeover automation provided in PLC for lifting the bogie automatically without any visual confirmation. By this, bogie changeover time has reduced to 33 sec. And now we are losing 2 green sheets for recycling.



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Green Sheets Loss		
	Before	After
Sheet Formation Time	16 Sec	16 Sec
Bogie Changeover time	45 sec	33 sec
Green Sheets Loss in No's	3 sheets	2 sheets
On Average Production of 300 MT/Day	72 Bogies Changeovers/Day	72 Bogies Changeovers/Day
Green Sheets Losses/Day	216 sheets	144 sheets
Production loss in terms of MT/Day	13.59 MT	9.06 MT
Total Saving of Production in MT/Day	4.526 M.T/Day	



Qualitative: Green sheets loss from machine has been reduced during piling bogie changeover. By this productivity is increased by reducing greens sheet loss to recycling unit.

Quantitative:

Productivity : Green sheet loss saving of **4.53 MT/Day**.

Quality : Quality of product increased due to reduction in usage of recycled sheets.

Cost : Cost saving of nearly **67 Lakhs/Year**.

Delivery : Machine production capacity has increased by **1454 MT/Year**.

Safety : System is safe with automation provided. No human interference is required.

Morale : By increase in production & safe working, employee morale has increased.

- Training on Updated Standard Operating Procedures are given to all the workmen for regular inspection, maintenance & to work safely.

	EVEREST INDUSTRIES LIMITED PODANUR IS/ISO 9001 : 2015 STANDARD OPERATING PROCEDURE	DOC No : EIL/SOP/26 REV. No : 00 PAGE : 1 OF 1 DATE : 15/10/2019
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Standard Operating Procedure: Piling Bogie Changeover

- 1) Align empty bogie position behind the bogie which is getting piled.
- 2) Once the bogie is piled fully, changeover the bogie.
- 3) When the new empty bogie is going to takes place it's position in piling machine, bogie oscillations are restricted by self actuating mechanical lock.]
- 4) Immediately when the new bogie is in position, with the help of PLC program modification, bogie is lifted automatically.
- 5) The entire lifting process is controlled by PLC without any human intervention.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection & Maintenance.
HOW TO DO	Regular Training on Standardized work instructions
FREQUENCY	Weekly

Horizontal Deployment:

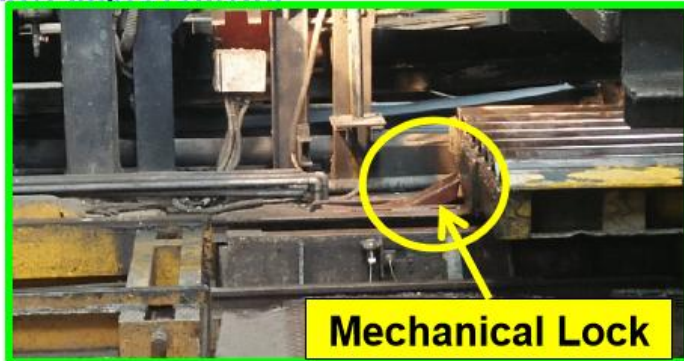
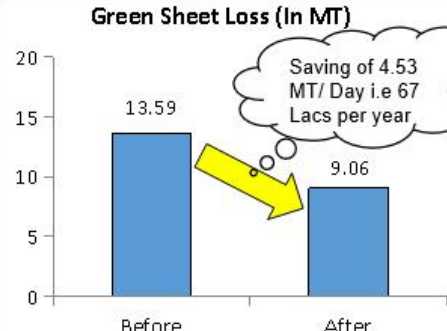
BW	CW	KW	LW	PW1	PW2	SW



- Under Progress



- Deployed

Kaizen Sheet			Category: Innovative	Company	MM/YY	S.No												
Reactive	Proactive	Innovative ✓		everest	10/19	01												
Kaizen Title: To reduce Green sheets loss and avoid recycling by reducing Piling bogie changeover time.				Implemented Area: PW1 Shop Floor														
Problem/Present Status: Green sheets loss during Piling bogie change as bogie takes long time to take position during changeover.		Before Improvement:  No Mechanical Lock		Implemented by: C.Murugesan, K.Karthikeyan, M.Vijay kumar, Y.Kesavaraj, M.Madhan Raj, Devaraj														
Root Cause Identification: Problem - Green sheets loss during Piling mixed bogie changeover. Why 1 - Piling mixed bogie takes time to take position at piling. Why 2 - Piling mixed bogie is oscillating back and fro while changeover and causing delay. Why 3 - There is no Locking provision to stop the bogie movement immediately after bogie changeover.		After Improvement:  Mechanical Lock		Result/Benefit: Productivity: Green sheets loss saving of 4.53 MT/Day. Cost : Cost saving of nearly INR.67 Lakhs/Year. Delivery: Machine production capacity has increased by 1454 MT/Year..														
Root cause		There is No Locking provision to stop the bogie movement immediately.		Green Sheet Loss (In MT) <table><caption>Green Sheet Loss (In MT)</caption><tr><th>Period</th><th>Loss (MT)</th></tr><tr><td>Before</td><td>13.59</td></tr><tr><td>After</td><td>9.06</td></tr></table>			Period	Loss (MT)	Before	13.59	After	9.06						
Period	Loss (MT)																	
Before	13.59																	
After	9.06																	
Idea to eliminate root cause		Provide Locking provision in piling bogie to avoid back & fro movement immediately while bogie changeover. Automation to be provided in PLC for lifting bogie.		Standardization: Training on SOP & Preventive maintenance as per schedule.														
Counter-measure		Mechanical lock provided in piling mixed bogie. Due to this bogie is locked without oscillating back & fro. Automation provided in PLC for lifting bogie.		Horizontal Deployment <table><tr><th>BW</th><th>CW</th><th>KW</th><th>LW</th><th>PW</th><th>SW</th></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>			BW	CW	KW	LW	PW	SW						
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